

A Multi-Dimensional Explainable Machine Learning Framework for Student Stress Analysis and Personalized Intervention Recommendation

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Abstract

This report documents the design, implementation, and evaluation of a complete end-to-end machine learning pipeline for student stress prediction, explanation, and personalized intervention. Starting from 1,100 student survey records with 20 numeric features and a three-class target variable (Low, Medium, and High stress), twelve classifiers were trained and compared under identical evaluation conditions. The top three models were selected and hyperparameter-tuned using RandomizedSearchCV. The tuned Extra Trees classifier achieved the best test accuracy of 89.55% and was selected as the final model. SHAP (SHapley Additive exPlanations) was applied to produce per-student feature contributions, which were aggregated into five interpretable stress dimensions — Academic, Psychological, Physical, Social, and Environmental. The dominant dimension for each student was used to generate personalized, rule-based intervention recommendations. A critical semantic refinement was implemented: for Low Stress students the system reports the “Dominant Low-Stress Explanation Dimension” rather than implying a stress-causing area, and only factors within the dominant dimension are displayed.

Keywords: student stress prediction, Extra Trees, SHAP explainability, Randomized-SearchCV, stress dimension detection, recommendation engine, multi-class classification

Contents

1	Introduction	3
2	Literature Review	3
3	Methodology	9
3.1	Dataset Description	11
3.2	Exploratory Data Analysis (EDA)	11
3.3	Data Preprocessing	13
3.3.1	Steps Applied	13
3.4	Model Training and Hyperparameter Tuning	14
3.5	SHAP Explainability and Dimension Aggregation	14
3.6	Personalized Recommendation Engine	15
4	Experiment Results	15
4.1	Baseline Comparative Evaluation of All Twelve Models	15
4.2	Hyperparameter Tuning Results	18
4.3	Final Model Evaluation	19
4.4	Feature Importance	21
4.5	SHAP Explainability Analysis	22
4.5.1	SHAP Bar Plot and Mean Absolute Feature Importance	22
4.5.2	SHAP Summary Plot	24
4.6	Multi-Dimensional Stress Profiles and Student Reports	26
5	Conclusion	28
5.1	Summary of Work	28
5.2	Final Decision and Suggestions	28
5.3	Limitations	29
5.4	Future Research Directions	29

1 Introduction

Student stress is a persistent and multi-dimensional problem in higher education, with documented effects on academic performance, physical health, and overall psychological well-being. While stress is a complex and multi-causal phenomenon, most existing computational studies on student stress reduce the problem to a single predictive task: given a set of survey responses, predict whether a student is experiencing low, medium, or high stress. Although such models can achieve reasonably high accuracy, they typically function as ‘black boxes’ that offer little insight into why a particular student has been classified into a given stress category, and consequently provide no actionable guidance for educators, counsellors, or the students themselves.

This report documents the design, implementation, and evaluation of a machine learning framework that goes beyond simple stress-level prediction. The framework is built around the Student Stress Factors Dataset, which contains 1,100 student records described by 20 numerical features spanning psychological, physiological, social, environmental, and academic factors, together with a three-level target variable, `stress_level` (0 = Low, 1 = Medium, 2 = High). The central aim of the project is to construct an explainable pipeline that: (1) predicts a student’s stress level using a comparative set of twelve machine learning algorithms, (2) identifies the best-performing model through evaluation, (3) applies SHAP to interpret individual and global model behaviour, (4) aggregates feature-level explanations into five interpretable stress dimensions, and (5) translates dimension-level insights into personalized, non-clinical intervention recommendations for each student.

What differentiates this project from many comparable studies is the combination of elements assembled in a single, connected pipeline: a twelve-model comparative baseline, honest reporting of hyperparameter tuning outcomes — including cases where tuning worsened performance — SHAP-based per-student explainability, five-dimensional stress profiling, and a corrected semantic interpretation for Low Stress predictions. The dominant-dimension filtering mechanism further ensures that only the top factors within the student’s dominant dimension are displayed, preventing cross-dimension confusion in the report output.

2 Literature Review

In a research study, Ahuja and Banga [1] applied machine learning algorithms to detect mental stress among university students. The study used a Perceived Stress Scale (PSS) questionnaire-based dataset collected from 206 students at Chitkara University, India. Four ML models were trained and evaluated: Random Forest, Naïve Bayes, SVM, and KNN. The SVM classifier achieved the highest accuracy of 85.71%, outperforming Random Forest (82.14%), Naïve

Bayes (71.43%), and KNN (67.86%). Although the study was an early and notable contribution to student stress detection, it was limited by its small dataset size and the absence of any model interpretability or explainability techniques.

In another study, Ratul et al. [2] analyzed the perceived psychological and social stress of 444 university students during the COVID-19 pandemic using a machine learning approach. They employed eight supervised ML algorithms—Decision Tree, Random Forest, Logistic Regression, KNN, Gradient Boosting, AdaBoost, XGBoost, and MLP—combined with PCA and Chi-squared feature reduction techniques. Their best-performing model, an MLP classifier optimized with PCA and Grid Search Cross-Validation, achieved an accuracy of 80.5%. The study emphasized the importance of hyperparameter optimization and feature engineering for improving model performance but was limited by convenience sampling bias and reliance on self-reported data.

Nath et al. [3] proposed a smart web interface for student mental health prediction using machine learning integrated with blockchain technology for data security. They collected data from both Kaggle (100 samples) and Bangladeshi universities (400 samples), totaling 500 student records. Twelve ML models were trained and compared, including Random Forest, Decision Tree, XGBoost, Logistic Regression, and SVM. The fine-tuned Random Forest model achieved the highest accuracy of 95.6%. Additionally, a Streamlit-based web application was built for real-time predictions, and blockchain was utilized to ensure data integrity. However, the study was limited to binary depression classification and used a relatively small dataset.

Singh et al. [4] conducted a comprehensive study on detecting mental stress in college students using machine learning algorithms. The study collected self-reported data from 843 students, which was validated by medical experts at AIIMS. Seven ML models were applied, including Decision Tree, Random Forest, SVM, AdaBoost, Naïve Bayes, Logistic Regression, and KNN. The SVM classifier achieved the highest accuracy of 95%, demonstrating the effectiveness of psychological questionnaire data for automated stress classification. Despite the high accuracy and large sample size, the study lacked explainability analysis, used only binary classification, and did not explore ensemble methods.

In a context-aware machine learning framework for stress monitoring, Ovi et al. [5] developed a comprehensive pipeline using two Kaggle datasets containing 1,100 and 843 student samples. They applied six base classifiers—SVM, Random Forest, Gradient Boosting, XGBoost, AdaBoost, and Bagging—along with five ensemble strategies including hard voting, soft voting, weighted voting, and stacking. Their stacking classifier achieved a remarkable accuracy of 99.53% on one dataset and 93.09% on the other, consistently outperforming individual classifiers. The study demonstrated the superiority of ensemble approaches for multi-class stress classification but lacked SHAP-based explainability and relied solely on survey-based data without real-time monitoring capabilities.

Ding et al. [6] proposed a systematic hybrid machine learning approach for stress prediction using physiological data. The dataset comprised 630 records with features such as snoring range, body temperature, respiration rate, limb movement, heart rate, and sleeping hours. They compared Logistic Regression, SVM, Random Forest, Gradient Boosting Machine (GBM), and AdaBoost, and then developed a hybrid model combining GBM and Random Forest through soft voting. The hybrid model achieved a perfect accuracy of 100%, validated by statistical T-tests. However, the result raises concerns about potential overfitting, particularly given the small dataset size and the exclusive use of physiological features without academic or psychological factors.

Chen and Lee [7] designed a novel stress analysis study using deep learning models applied to physiological signals collected during Sudoku puzzle-solving tasks. Their dataset consisted of PPG, ECG, and EEG signals from 30 university participants. Three deep learning models—Long-term Recurrent Convolutional Network (LRCN), Self-supervised CNN, and StressNeXt—were evaluated. The LRCN model achieved the highest accuracy of 98.78%. While the study demonstrated the strong potential of deep learning on physiological data, the very small sample size of 30 participants and the requirement for expensive wearable sensors significantly limit the scalability and practical applicability of this approach.

A group of researchers led by Ciharova et al. [8] investigated machine-learning detection of stress severity on a continuous scale using multimodal data including visual (facial), acoustic (voice), verbal, and physiological (ECG, EDA) features. The study involved 42 university students subjected to the Trier Social Stress Test (TSST). Bayesian Ridge Regression and Logistic Regression models were employed. For continuous stress prediction, the algorithm achieved a weak association with an R^2 of only 0.154, while for binary classification the accuracy reached approximately 71%. This study was the first to attempt continuous-scale stress severity detection using multimodal data, but it was constrained by the very small sample size and its lab-only experimental setting.

Nnadi et al. [9] explored the prediction of students' adaptability levels using Explainable AI in educational machine learning models. Their study utilized a dataset of 1,205 instances and employed several ML algorithms, with Random Forest demonstrating the highest accuracy at 91%. The study uniquely incorporated five XAI techniques—SHAP, LIME, Anchors, ALE, and Counterfactual explanations—to interpret the model's predictions. SHAP values highlighted "Class Duration" as the most significant feature (mean SHAP value: 0.175), while Anchors provided high-confidence rule-based explanations with 97.32% confidence. This study demonstrated the value of combining multiple explainability techniques in educational ML applications, though it focused on student adaptability rather than stress prediction.

Abdul Rahman et al. [10] investigated machine learning-based prediction of mental well-being using health behavior data from university students across 17 universities in Southeast Asia.

They compared multiple ML algorithms including generalized linear models, k-nearest neighbor, naïve Bayes, neural networks, random forest, recursive partitioning, bagging, and boosting. Random Forest and adaptive boosting algorithms achieved the highest accuracy for identifying negative mental well-being traits. The study identified the number of sports activities per week, body mass index, GPA, sedentary hours, and age as the top five most salient features. This multi-site cross-sectional study was one of the largest in the domain but lacked explainability techniques and did not address stress specifically.

In a study by Ramasamy et al. [11], a student mental health prediction model was developed using a combination of Fuzzy Inference Systems and machine learning. The study surveyed 1,182 students from educational institutions in Delhi NCR, India. They employed Mamdani Fuzzy Inference with Min-Max labeling for stress level categorization, and applied Pearson correlation for feature selection. MLR, MLP, and D14jMlpClassifier were used as prediction models, with MLP-based models achieving the highest accuracy. The study's unique contribution was the integration of fuzzy logic with machine learning, but it was limited by complex methodology and absence of model explainability.

El-Morr et al. [12] developed predictive machine learning models for assessing depression, anxiety, and stress among 261 Lebanese university students. They trained seven classifiers—Decision Tree, Random Forest, KNN, Naïve Bayes, Logistic Regression, SVM, and ANN—to separately predict each mental health condition using the DASS-21 scale. Random Forest achieved the highest accuracy of 89% for depression prediction. This cross-cultural study provided important insights into mental health prediction in Middle Eastern educational contexts, but was constrained by a small regional sample, the absence of ensemble methods, and no model explainability.

Barbayannis et al. [13] conducted a comprehensive survey-based study examining the relationship between academic stress and mental well-being in 843 college students. The study employed purely statistical methods, including correlation analysis and regression models, to investigate how academic workload and the COVID-19 pandemic impacted student well-being. The findings established significant correlations between academic stress levels and mental health indicators. However, the study did not employ any machine learning models, making it a foundational reference for understanding stress factors rather than a predictive tool.

Macalli et al. [14] employed a machine learning approach for predicting suicidal thoughts and behaviours among college students using data from the French i-Share cohort study. The longitudinal study included 5,066 students and used random forest models with 70 potential predictors including sociodemographic characteristics, mental health indicators, and substance use patterns. The model achieved an AUC of 0.80, sensitivity of 79% for girls and 81% for boys, and positive predictive value of 40% for girls and 36% for boys. Among all 70 predictors, four showed the highest predictive power: 12-month suicidal thoughts, trait anxiety, depression

symptoms, and self-esteem. This large-scale longitudinal study demonstrated the feasibility of ML-based screening in student populations but focused exclusively on suicidal behaviours rather than general stress prediction.

Liu and Yu [15] proposed a students' stress prediction and explainable analysis model based on improved decision trees. They compared nine machine learning algorithms including Logistic Regression, Decision Tree, SVM, KNN, Random Forest, XGBoost, LightGBM, Gradient Boosting, and Naïve Bayes to identify the optimal base model. The standard Decision Tree achieved an accuracy of 0.909. They then introduced the Harris Hawks Optimization (HHO) algorithm to optimize the Decision Tree model, which improved the accuracy to 0.927 with the fewest misclassified samples. Additionally, SHAP analysis was applied to interpret the predictions, revealing that blood pressure, social support, and depression were the key features affecting stress level predictions. This study provides a novel contribution through the combination of metaheuristic optimization with decision trees and SHAP explainability.

De Filippis and Foysal [16] conducted a comprehensive analysis of stress factors affecting students using a machine learning approach. Their study utilized the StressLevelDataset containing records from 1,100 students across various educational institutions. They employed correlation analysis and feature importance analysis using Random Forest models to identify and rank the most significant stressors. The findings suggested that psychological factors such as self-esteem and physiological factors like sleep quality are the most crucial predictors of stress levels. A significant negative correlation was observed between anxiety levels and academic performance. While the factor analysis was thorough, the study was limited in its model comparison, relying primarily on Random Forest, and lacked ensemble methods and explainability techniques.

Shekhar and Kumar [17] presented a framework for predicting mental stress levels among students using machine learning and purely academic data. Their consolidated dataset included examination scores, attendance proportions, and reported study durations. They trained Logistic Regression, SVM, Random Forest, and Gradient Boosting models to classify students into Low, Medium, or High stress groups. The strongest model reached accuracy in the 60–65% range, indicating that while academic variables can signal underlying stress conditions, they are insufficient on their own for reliable prediction. The study highlights the need for supplementary features from lifestyle, psychological, and social domains for accurate stress classification.

Anand et al. [18] focused on enhancing diagnostic decision-making through ensemble learning for stress classification. Using the Kaggle Student Stress Factors dataset of 1,100 students containing 20 features across psychological, physiological, environmental, academic, and social categories, they evaluated seven classifiers—Random Forest, Gradient Boosting, XGBoost, AdaBoost, Bagging, Voting, and Stacking. Their Stacking classifier achieved the highest accu-

racy of 92.41%. The study demonstrated the effectiveness of ensemble learning strategies for stress classification but did not incorporate any explainability analysis or feature importance ranking.

Doma et al. [19] proposed a classification system for student stress levels using a hybrid machine learning model. Working with the same Kaggle dataset of 1,100 students, they developed an ensemble approach combining multiple classifiers. Their hybrid model achieved an accuracy of 92.41%, matching the results of Anand et al. [18] on the same dataset. While the study confirmed the reliability of ensemble approaches for multi-class stress prediction, it offered limited methodological novelty and did not include explainability techniques.

Tariq et al. [20] published a study on explainable artificial intelligence for predictive modeling of student stress in higher education. The research applied six classification algorithms Logistic Regression, SVM, Decision Tree, Random Forest, Gradient Boosting, and XGBoost optimized using grid search and cross-validation. The Random Forest model achieved the highest classification accuracy of 89%, followed by XGBoost at 87% and Gradient Boosting at 85%. SHAP analysis was conducted to interpret model predictions, revealing five principal predictors: blood pressure, perceived safety, sleep quality, teacher-student relationship, and participation in extracurricular activities. The study further applied game theory concepts to derive actionable intervention strategies for educational institutions. While comprehensive in its use of SHAP and game theory, the study was limited to a single dataset without multi-dataset validation.

It can be seen that a wide variety of machine learning models, ensemble techniques, and deep learning architectures have been employed to address the challenge of predicting student mental stress, which remains a pressing concern in academic institutions worldwide. Studies utilizing ensemble methods such as stacking and voting classifiers have generally produced superior results compared to individual classifiers. However, a majority of the reviewed studies operate as “black-box” models that provide no insight into why a particular student is predicted to be stressed. Only a handful of studies incorporate explainability techniques such as SHAP, and none have comprehensively combined multi-class stress prediction using ensemble ML models with SHAP-based explainability on a student stress dataset containing both academic and behavioral features. Based on this literature review, the researchers decided to develop an Explainable Machine Learning Framework that integrates high-accuracy ensemble models including Random Forest, XGBoost, and LightGBM with SHAP-based explainability for multi-class student stress prediction, thereby addressing the identified gap in the literature. Table 1 provides a summary comparison of the reviewed studies.

Table 1: Summary of reviewed literature on machine learning for student stress prediction.

Study	Year	Best Model	Accuracy	XAI	Dataset
Ahuja & Banga	2019	SVM	85.71%	No	206
Ratul et al.	2023	MLP + PCA	80.50%	No	444
Nath et al.	2025	Random Forest	95.60%	No	500
Singh et al.	2024	SVM	95.00%	No	843
Ovi et al.	2025	Stacking	99.53%	No	1,100
Ding et al.	2023	Hybrid GBM+RF	100%	No	630
Chen & Lee	2023	LRCN	98.78%	No	30
Ciharova et al.	2025	Bayesian Ridge	71%	No	42
Nnadi et al.	2024	Random Forest	91.00%	SHAP+LIME	1,205
Abdul Rahman et al.	2023	RF + AdaBoost	—	No	Multi-site
Ramasamy et al.	2024	MLP + Fuzzy	—	No	1,182
El-Morr et al.	2024	Random Forest	89.00%	No	261
Barbayannis et al.	2022	Statistical	—	No	843
Macalli et al.	2021	Random Forest	AUC 0.80	No	5,066
Liu & Yu	2026	DT + HHO	92.70%	SHAP	1,100
De Filippis & Foysal	2024	Random Forest	—	No	1,100
Shekhar & Kumar	2025	Gradient Boost.	65%	No	Academic
Anand et al.	2024	Stacking	92.41%	No	1,100
Doma et al.	2025	Hybrid Ensemble	92.41%	No	1,100
Tariq et al.	2025	Random Forest	89.00%	SHAP	1,100
This Study	2026	Extra Trees	89.55%	SHAP	1,100

3 Methodology

This section describes the complete pipeline in the order in which it was implemented: from dataset loading and exploratory analysis through preprocessing, model training and hyperparameter tuning, SHAP explainability, five-dimensional stress profiling, and personalized recommendation generation. All code was written in Python 3 and executed in Google Colab. The primary libraries used were Pandas and NumPy for data manipulation, Matplotlib and Seaborn for visualization, scikit-learn for preprocessing and models, XGBoost and LightGBM for gradient-boosting variants, and SHAP for post-hoc model explanation.

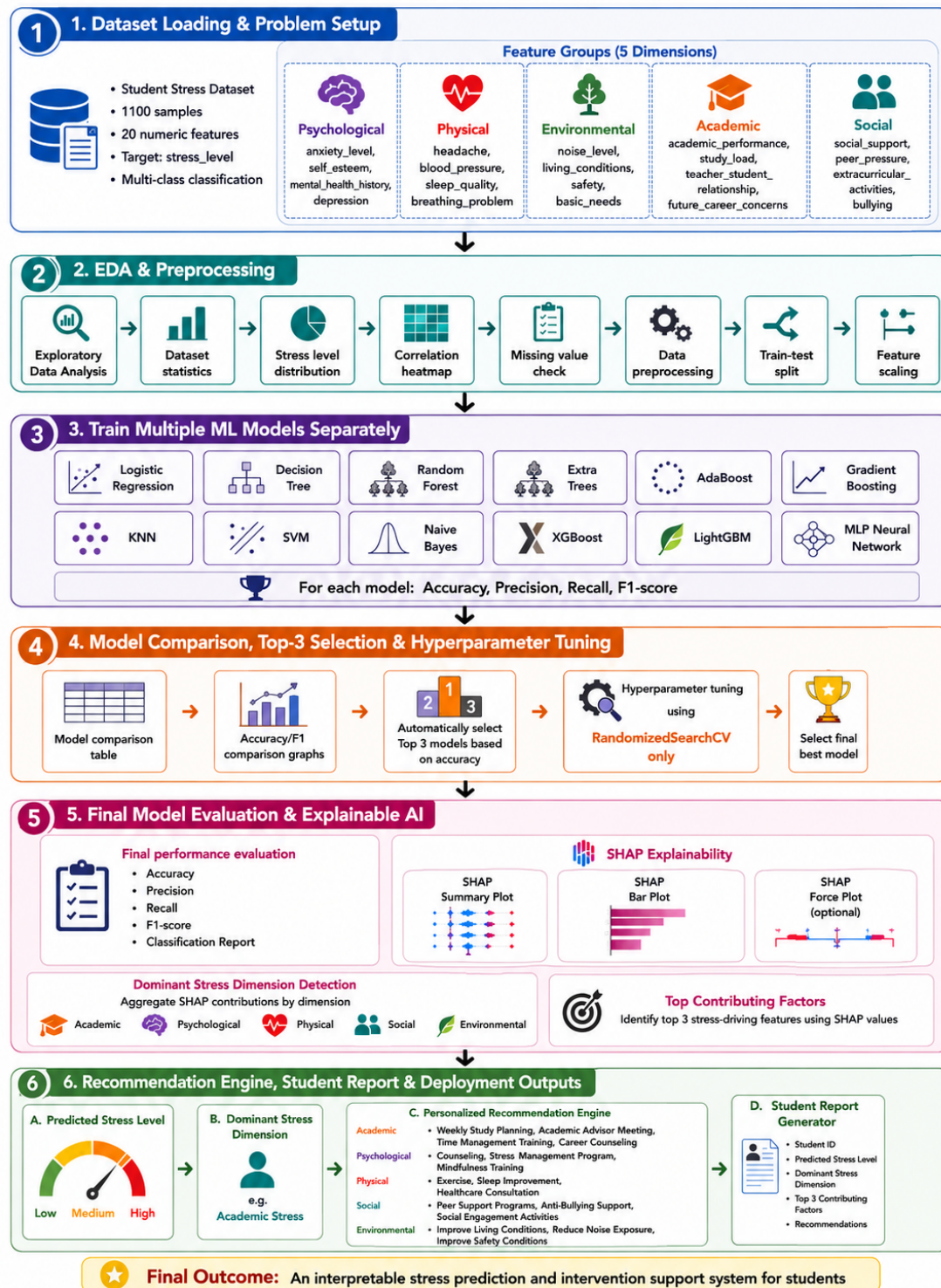


Figure 1: Complete pipeline workflow: from dataset loading and EDA through preprocessing, model training, hyperparameter tuning, SHAP explainability, dimension profiling, and personalized recommendation generation.

3.1 Dataset Description

The dataset used throughout this project is the Student Stress Factors Dataset, obtained from Kaggle. It contains 1,100 rows and 21 columns: 20 numeric input features and one target column named `stress_level`, encoded as 0 (Low Stress), 1 (Medium Stress), or 2 (High Stress). Every column is integer-valued, which eliminates the need for categorical encoding. The 20 input features were divided into five conceptual stress dimensions (Table 2) for use at the explanation stage only — this mapping plays no role in model training.

Table 2: Mapping of the 20 input features to five interpretable stress dimensions.

Dimension	Features (4 per dimension)
Academic Stress	study_load, academic_performance, teacher_student_relationship, future_career_concerns
Psychological Stress	anxiety_level, self_esteem, mental_health_history, depression
Physical Stress	headache, blood_pressure, sleep_quality, breathing_problem
Social Stress	social_support, peer_pressure, extracurricular_activities, bullying
Environmental Stress	noise_level, living_conditions, safety, basic_needs

3.2 Exploratory Data Analysis (EDA)

Before any modelling decisions were made, an exploratory data analysis (EDA) was conducted to understand the structure, quality, and distributional properties of the dataset. The analysis proceeded through four steps: a structural check, a missingness audit, a class distribution review, and a correlation analysis.

The missingness audit confirmed that zero missing values are present across all 21 columns, as shown in Figure 2. This eliminates any need for imputation strategies.

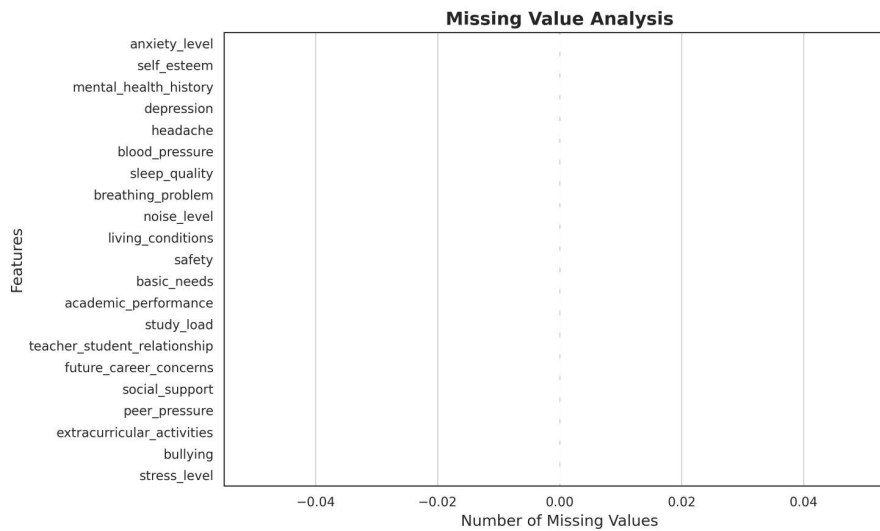


Figure 2: Missing-value count for each of the 21 dataset columns. Every bar registers zero, confirming that no imputation step was required.

The class distribution was examined next. Figure 3 shows that the three target classes are nearly balanced: 373 records are Low Stress (33.9%), 358 are Medium Stress (32.5%), and 369 are High Stress (33.5%). This near-balance confirms that class-rebalancing techniques such as SMOTE are unnecessary.

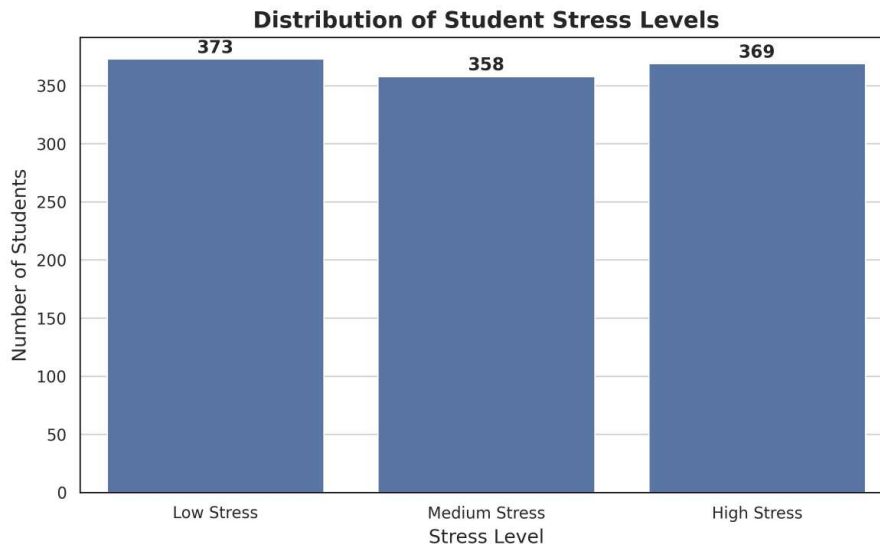


Figure 3: Distribution of the three stress-level classes (Low: 373, Medium: 358, High: 369). The near-equal distribution confirms the dataset is balanced.

The correlation analysis produced a Pearson correlation heatmap (Figure 4). `blood_pressure` shows one of the strongest positive correlations with the `stress_level` target. `self_esteem` and `sleep_quality` show the strongest negative correlations.

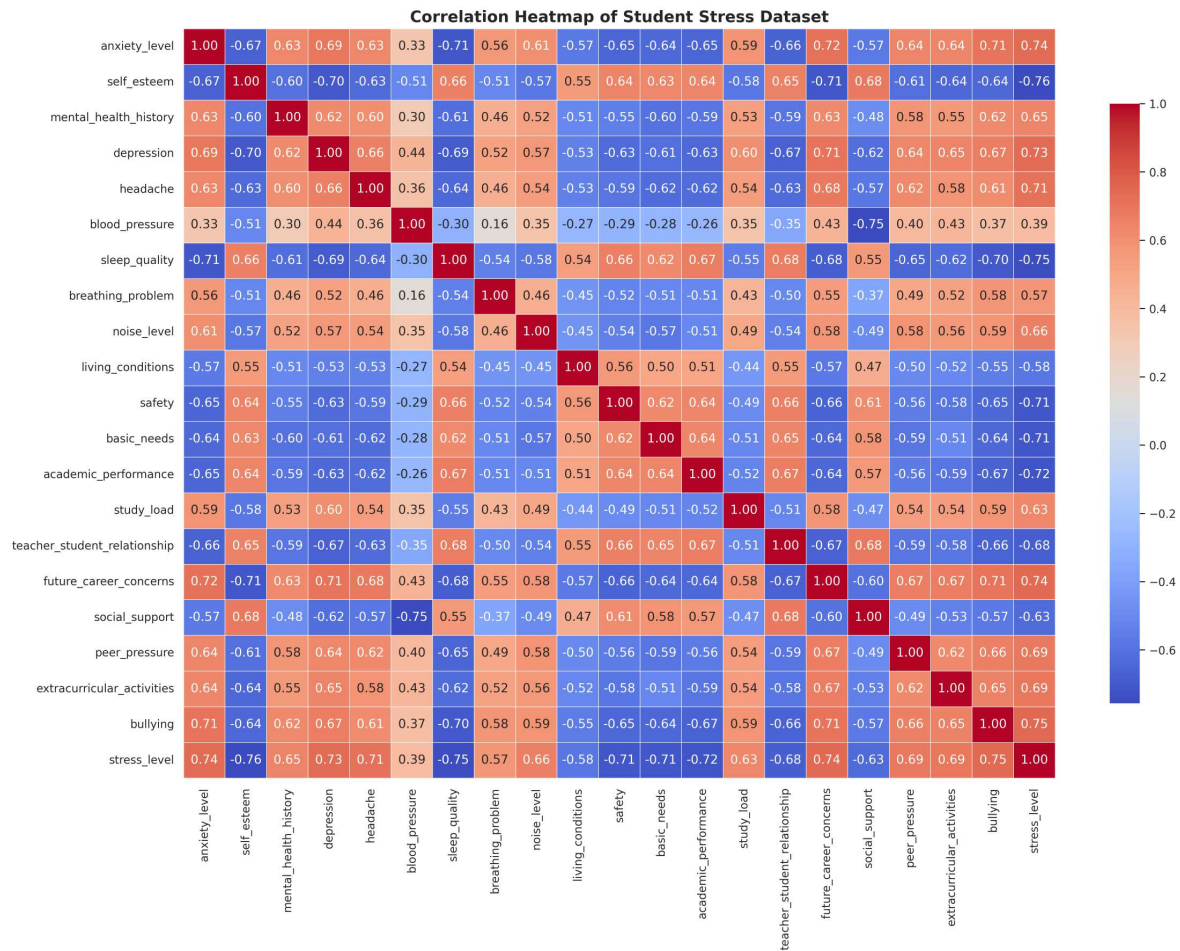


Figure 4: Pearson correlation heatmap for all 20 input features and the target variable (`stress_level`). Warm colours indicate positive correlation; cool colours indicate negative correlation with stress.

3.3 Data Preprocessing

3.3.1 Steps Applied

Based on the EDA findings, the minimum necessary preprocessing consisted of three steps: feature-target separation, stratified train-test splitting, and feature scaling. The dataset was separated into X (20 input features) and y (`stress_level`). An 80/20 stratified split was applied, producing 880 training records and 220 test records while preserving class proportions. `StandardScaler` was fitted exclusively on X_{train} and applied using `.transform()` on X_{test} — this leakage-avoidance ordering is critical to prevent the test distribution from influencing training normalization.

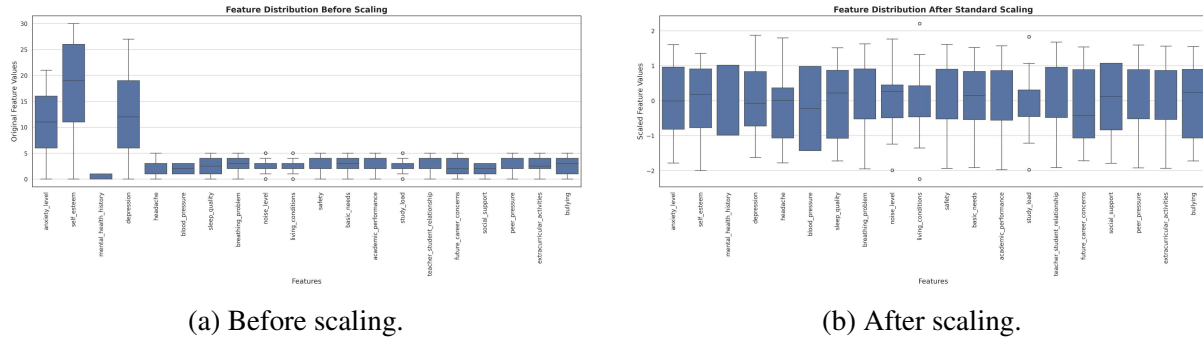


Figure 5: Feature value distributions before and after StandardScaler transformation.

3.4 Model Training and Hyperparameter Tuning

Twelve classifiers were selected to cover the principal families of supervised learning: Logistic Regression, Decision Tree, Random Forest, Extra Trees, AdaBoost, Gradient Boosting, KNN, SVM, Naive Bayes, XGBoost, LightGBM, and MLP Neural Network. Each model was trained and evaluated on the same 220-record held-out test set under identical conditions.

Rather than exhaustively tuning all models, an efficient filtering strategy was adopted: only the top three models by baseline accuracy were selected to proceed to hyperparameter tuning. For these three models, RandomizedSearchCV was applied with 5-fold stratified cross-validation and `n_iter=20`, sampling random configurations from the search spaces defined in Table 3.

Table 3: RandomizedSearchCV search spaces for the top three tuned models (`n_iter=20`, `cv=5`).

Model	Parameter	Search Space
Extra Trees	<code>n_estimators</code>	50–200 (uniform int)
Extra Trees	<code>max_depth</code>	None, 10–20
Extra Trees	<code>min_samples_split</code>	2–10 (uniform int)
Extra Trees	<code>min_samples_leaf</code>	1–4 (uniform int)
Random Forest	<code>n_estimators</code>	50–200 (uniform int)
Random Forest	<code>max_depth</code>	None, 10–20
Random Forest	<code>min_samples_split</code>	2–10 (uniform int)
Gradient Boosting	<code>n_estimators</code>	50–200 (uniform int)
Gradient Boosting	<code>learning_rate</code>	0.01–0.20 (uniform)

3.5 SHAP Explainability and Dimension Aggregation

Once the final model was selected, SHAP was applied to compute feature-level contribution scores for every test student. To aggregate the 20 feature-level SHAP values into the five stress

dimensions, the absolute SHAP values within each dimension were summed and normalized, producing a percentage contribution for each dimension. The dimension with the highest percentage is identified as the *Dominant Stress Dimension* for that student.

A critical semantic refinement was implemented for Low Stress students: the system reports a “*Dominant Low-Stress Explanation Dimension*” rather than implying a stress-causing area. Additionally, the system restricts the displayed top factors to only those features belonging to the student’s dominant dimension, preventing cross-dimension confusion.

3.6 Personalized Recommendation Engine

The final stage connects dimension-level insights to actionable, rule-based interventions (Table 4). For Low Stress predictions, no intervention is generated. For Medium and High Stress, the dominant dimension determines which recommendations are presented.

Table 4: Recommendation library used by the personalized intervention engine.

Dimension	Recommendations
Academic	Weekly Study Planning; Academic Advisor Meeting; Time Management Training; Career Counseling
Psychological	Counseling Referral; Stress Management Program; Mindfulness Training
Physical	Exercise Routine; Sleep Improvement Plan; Healthcare Consultation
Social	Peer Support Programs; Anti-Bullying Support; Social Engagement Activities
Environmental	Improve Living Conditions; Reduce Noise Exposure; Improve Safety Conditions

4 Experiment Results

This section presents the results from each stage of the pipeline in sequence: baseline comparative evaluation, hyperparameter tuning outcomes, final model detailed evaluation, SHAP-based feature importance analysis, multi-dimensional stress profiling, and student-level report examples.

4.1 Baseline Comparative Evaluation of All Twelve Models

Each of the twelve baseline classifiers was evaluated on the 220-record held-out test set. Table 5 presents the full baseline results. Figures 6, 7, and 8 provide visual comparisons.

Table 5: Baseline test-set performance of all twelve classifiers (sorted by descending accuracy).

Model	Accuracy	Precision	Recall	F1
Extra Trees	88.64%	88.63%	88.64%	88.60%
LightGBM	88.64%	88.64%	88.64%	88.60%
XGBoost	88.64%	88.79%	88.64%	88.64%
AdaBoost	88.64%	88.82%	88.64%	88.59%
Logistic Regression	88.18%	88.16%	88.18%	88.17%
Random Forest	88.18%	88.27%	88.18%	88.14%
Naive Bayes	88.18%	89.43%	88.18%	88.33%
SVM	87.73%	87.73%	87.73%	87.71%
Decision Tree	87.27%	87.27%	87.27%	87.26%
Gradient Boosting	87.27%	87.30%	87.27%	87.22%
KNN	87.27%	87.44%	87.27%	87.21%
MLP Neural Network	86.82%	86.94%	86.82%	86.77%

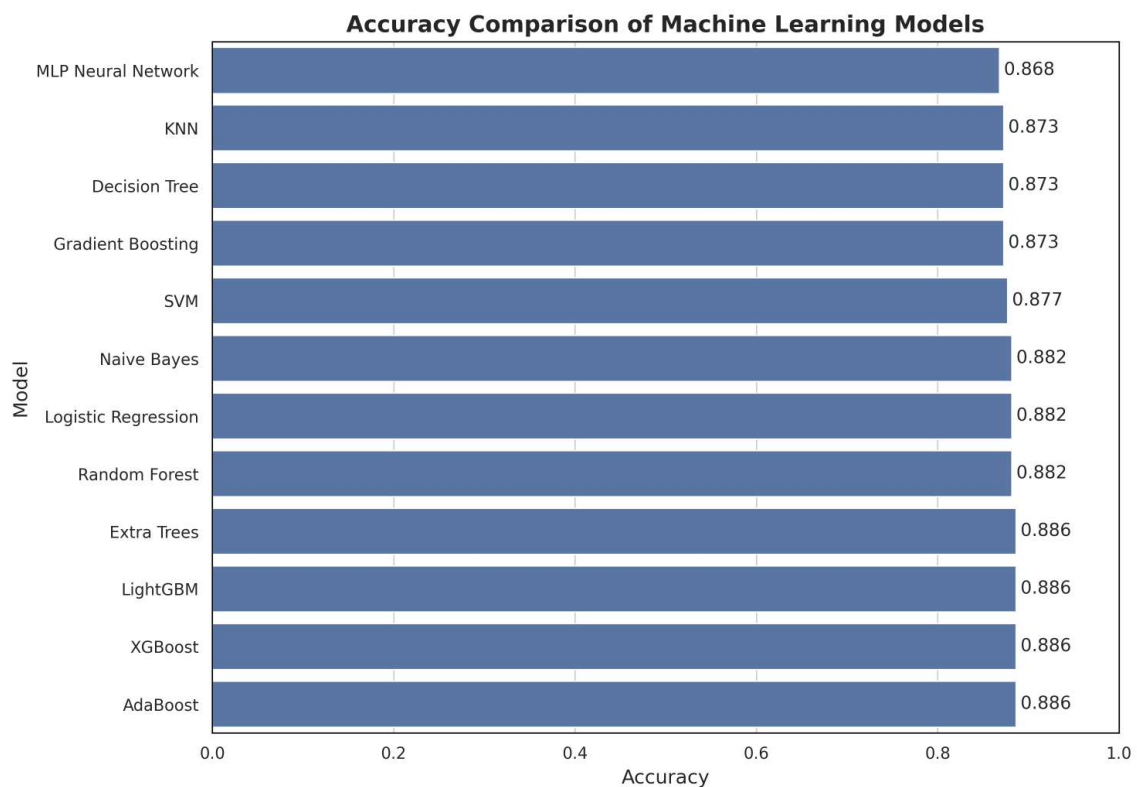


Figure 6: Test accuracy comparison of all twelve baseline models.

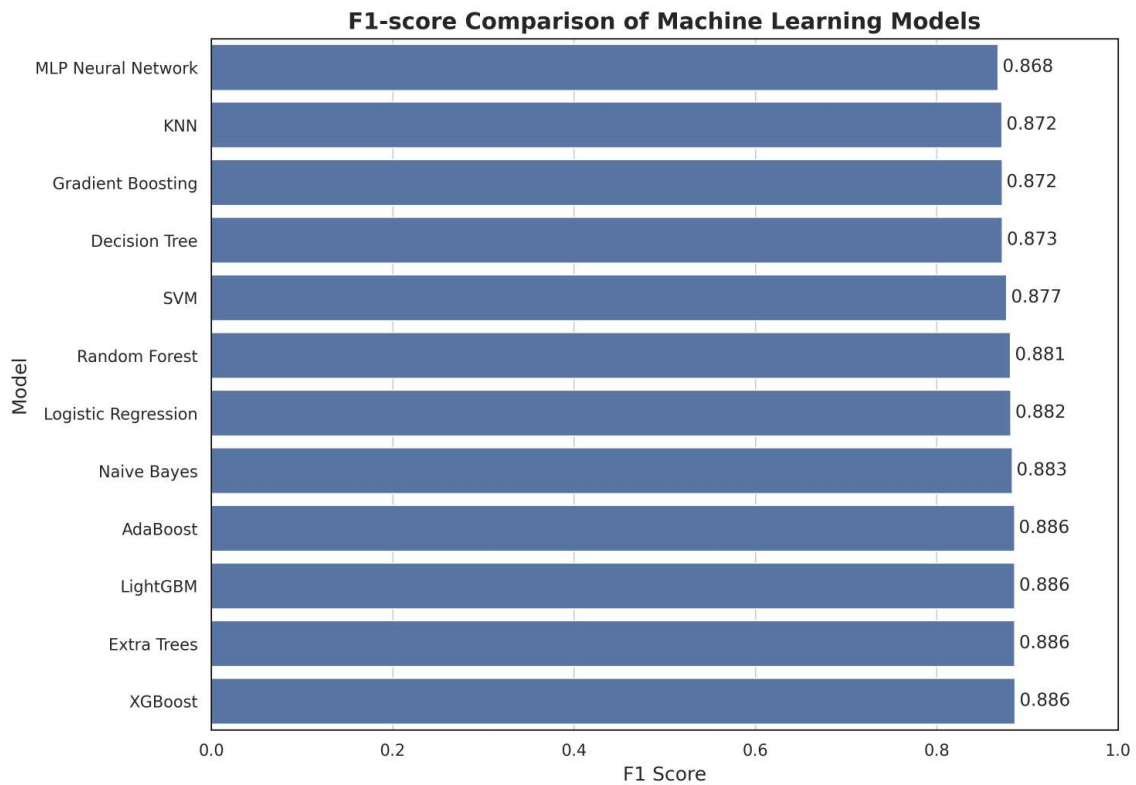


Figure 7: F1-score comparison of all twelve baseline models.

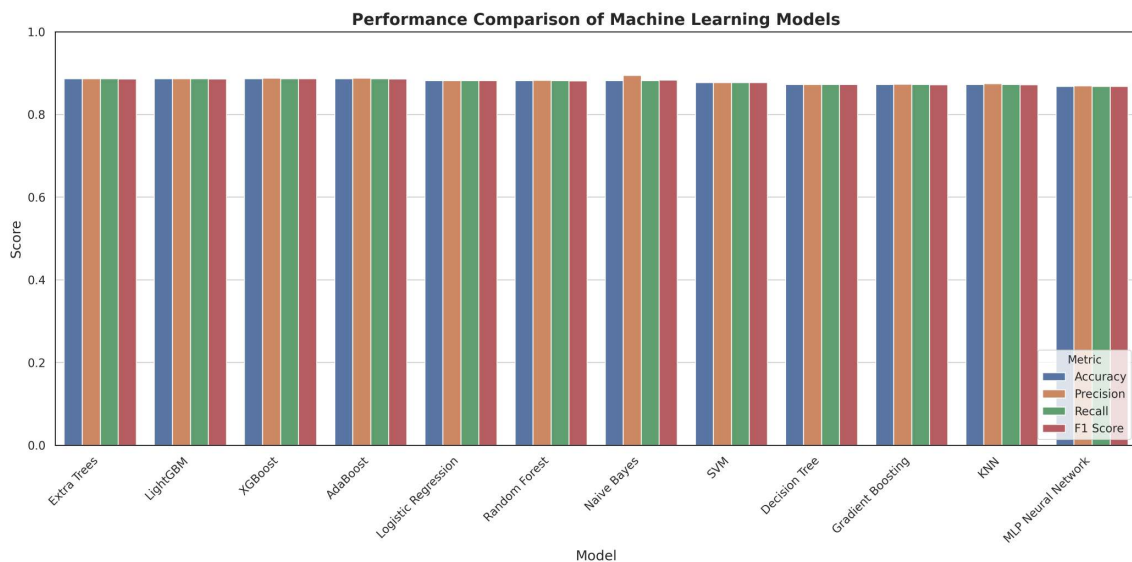


Figure 8: All-metrics comparison (Accuracy, Precision, Recall, F1-score) across all twelve models.

The spread in test accuracy across all twelve models is remarkably narrow — only 1.82 percentage points separate the best-performing models (Extra Trees, LightGBM, XGBoost, and AdaBoost at 88.64%) from the worst (MLP Neural Network at 86.82%). Based on baseline performance, Extra Trees, Random Forest, and Gradient Boosting were automatically selected

as the top three candidates for hyperparameter tuning.

4.2 Hyperparameter Tuning Results

RandomizedSearchCV was applied to the top three models. The results demonstrated a varied impact — an honest finding documented here rather than omitted. Figures 9, 10, and 11 show the before/after comparison.

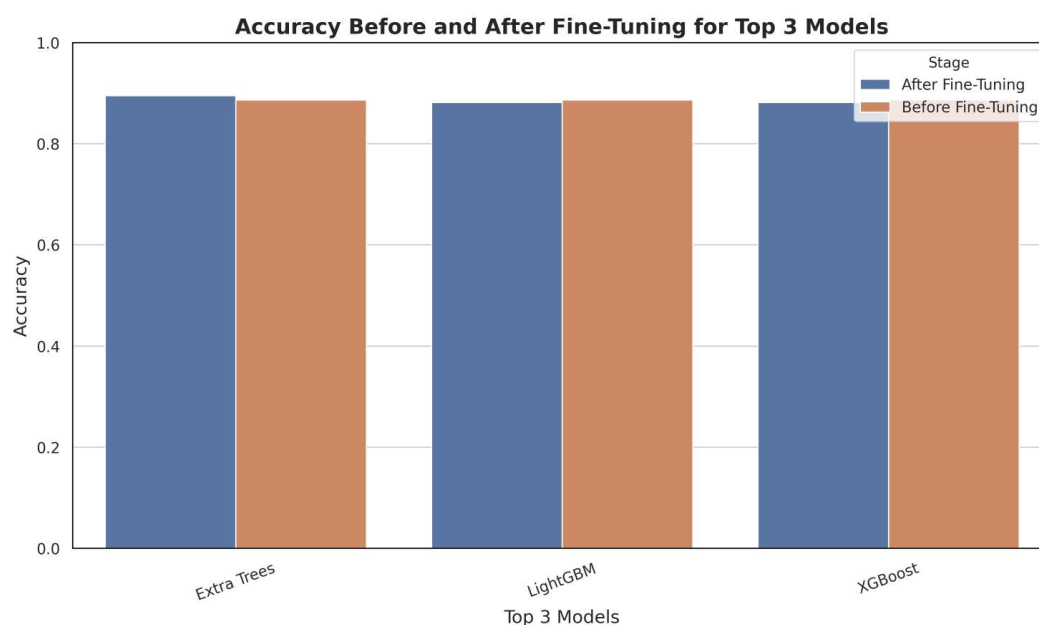


Figure 9: Accuracy before and after hyperparameter tuning for the top three models.

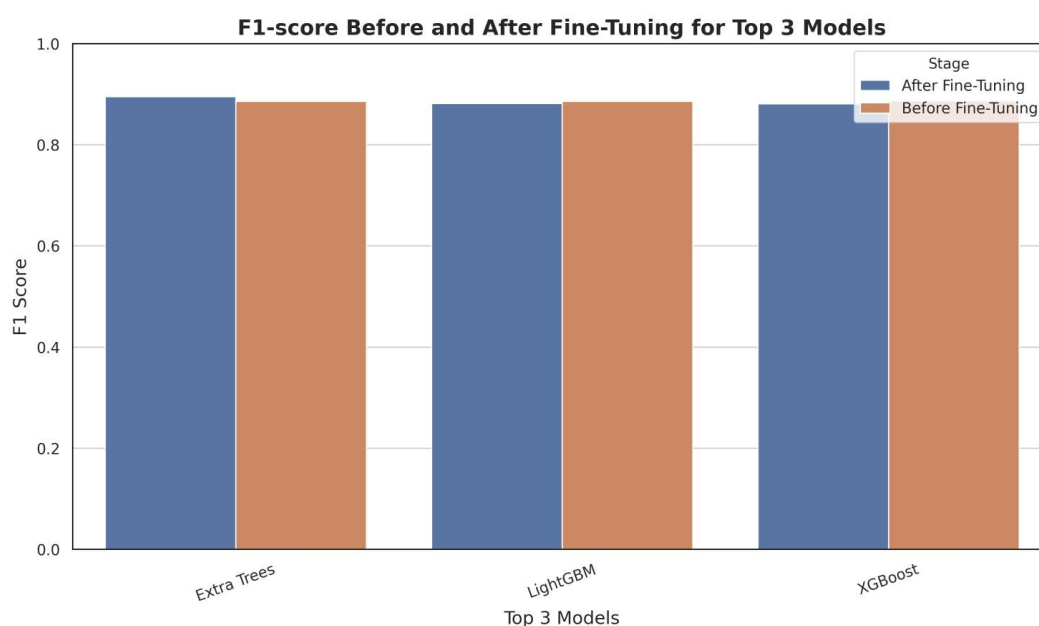


Figure 10: F1-score before and after hyperparameter tuning for the top three models.

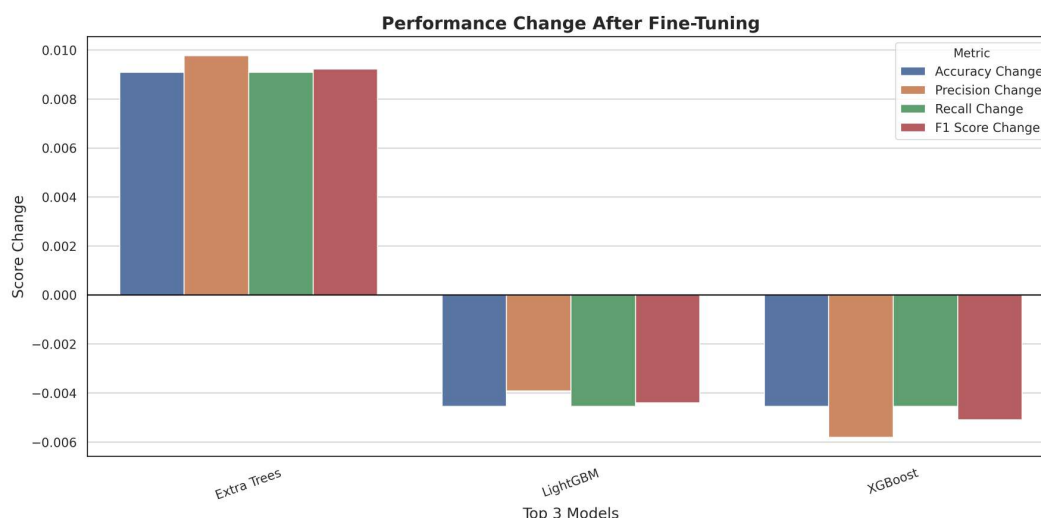


Figure 11: Performance change (Δ accuracy) after hyperparameter tuning for the top three models.

Tuning the Extra Trees classifier confirmed its robustness: it improved from a baseline accuracy of 88.64% to a final test accuracy of 89.55%, with an optimal configuration of 108 trees, maximum depth 17, minimum samples per split 8, minimum samples per leaf 1, and Gini criterion. A finding worth documenting explicitly is that the same RandomizedSearchCV procedure caused LightGBM and XGBoost to decline slightly — illustrating that for datasets with moderate sample sizes, aggressively tuning complex gradient-boosted methods with a limited search budget can sometimes lead to overfitting the cross-validation folds at the expense of generalization. Based on post-tuning test performance, **Extra Trees was selected as the final model.**

4.3 Final Model Evaluation

The detailed per-class classification report for the final tuned Extra Trees model is shown in Table 6. The confusion matrix and classification report heatmap are shown in Figures 12 and 13.

Table 6: Per-class classification report for the final Extra Trees model (220-record test set).

Class	Precision	Recall	F1-Score	Support
Low (0)	0.91	0.85	0.88	74
Medium (1)	0.91	0.93	0.92	72
High (2)	0.87	0.91	0.89	74
Accuracy (overall)	—	—	0.90	220
Macro Average	0.89	0.90	0.89	220
Weighted Average	0.90	0.90	0.90	220

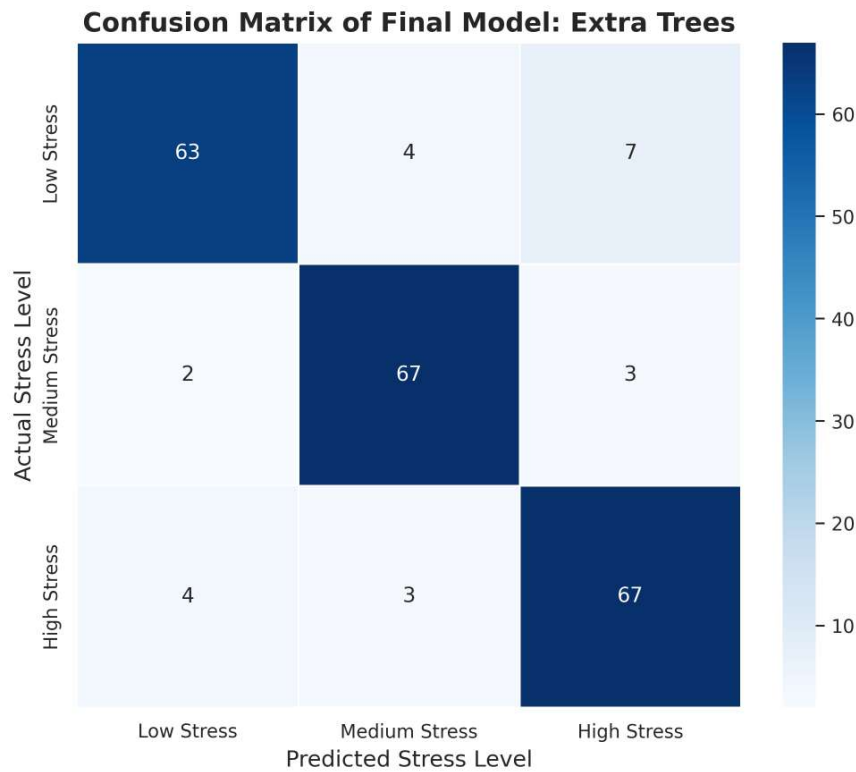


Figure 12: Confusion matrix for the final tuned Extra Trees classifier. Rows represent actual class; columns represent predicted class.

The confusion matrix shows that the Medium-stress class was classified most reliably, with 67 of 72 students correctly identified (recall = 0.93) and only 5 misclassified (2 as Low, 3 as High). The Low class was correctly identified for 63 of 74 students (recall = 0.85); the 11 misclassifications were split between Medium (4 students) and High (7 students), with the latter representing the most concerning error type, since it corresponds to a student experiencing low stress being flagged as high-stress. The High class achieved a recall of 0.91 (67 of 74 correct), with the remaining 7 misclassifications split between Low (6 students) and Medium (1 student). Overall, the model's precision for the High class (0.87) is slightly lower than its recall (0.91), indicating a modest tendency toward false positives for the High class — that is, the model is somewhat more likely to over-predict High stress than to miss a genuinely high-stress student, which, from a student-support perspective, is arguably the safer direction for errors to fall.

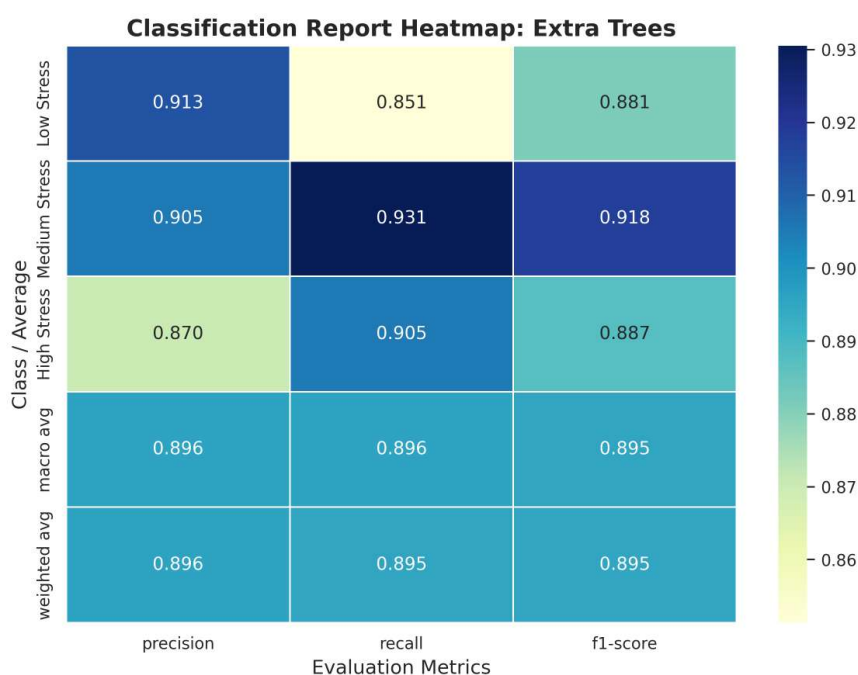


Figure 13: Classification report heatmap for the final Extra Trees model. Dark, uniform shades confirm balanced performance across all three classes.

Figure 13 visually corroborates the metrics in Table 6. The dark, uniform shades across the core metrics — particularly the recall scores of 0.85, 0.93, and 0.91 for Low, Medium, and High stress respectively — confirm that the Extra Trees classifier maintains its predictive strength uniformly, without sacrificing performance on any single class.

4.4 Feature Importance

Feature importance was generated for the final Extra Trees model to provide a global understanding of which variables primarily influence the stress level classification.

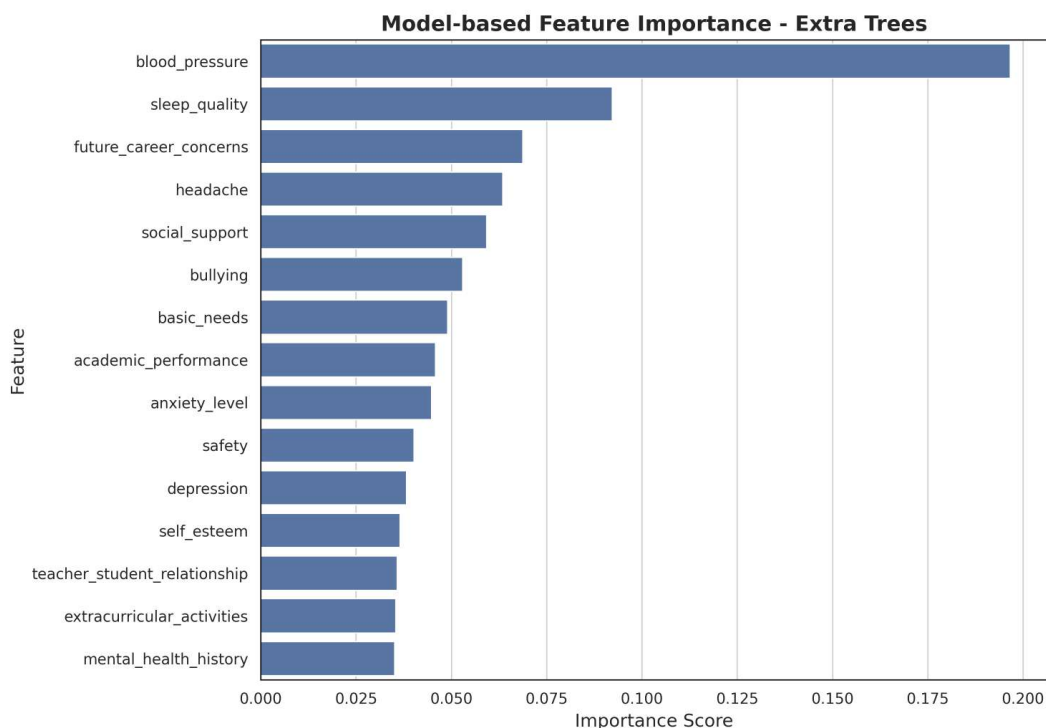


Figure 14: Feature importance plot for the final Extra Trees model.

The feature importance analysis reveals that physiological and physical features — particularly `blood_pressure`, `headache`, and `sleep_quality` — consistently emerge as the most influential variables. Academic features such as `academic_performance` and `future_career_concerns` also appear among the top contributors.

4.5 SHAP Explainability Analysis

To make the project highly explainable and research-oriented, SHAP (SHapley Additive exPlanations) was applied to explain the predictions of the final Extra Trees model. SHAP reveals how each feature contributes to the model output for every student, explaining exactly why a given student is predicted as having Low, Medium, or High Stress. Three SHAP visualizations were generated.

4.5.1 SHAP Bar Plot and Mean Absolute Feature Importance

Global feature importance was generated to show which features are most important for the model's predictions overall, providing a global understanding of which variables primarily influence the stress level classification.

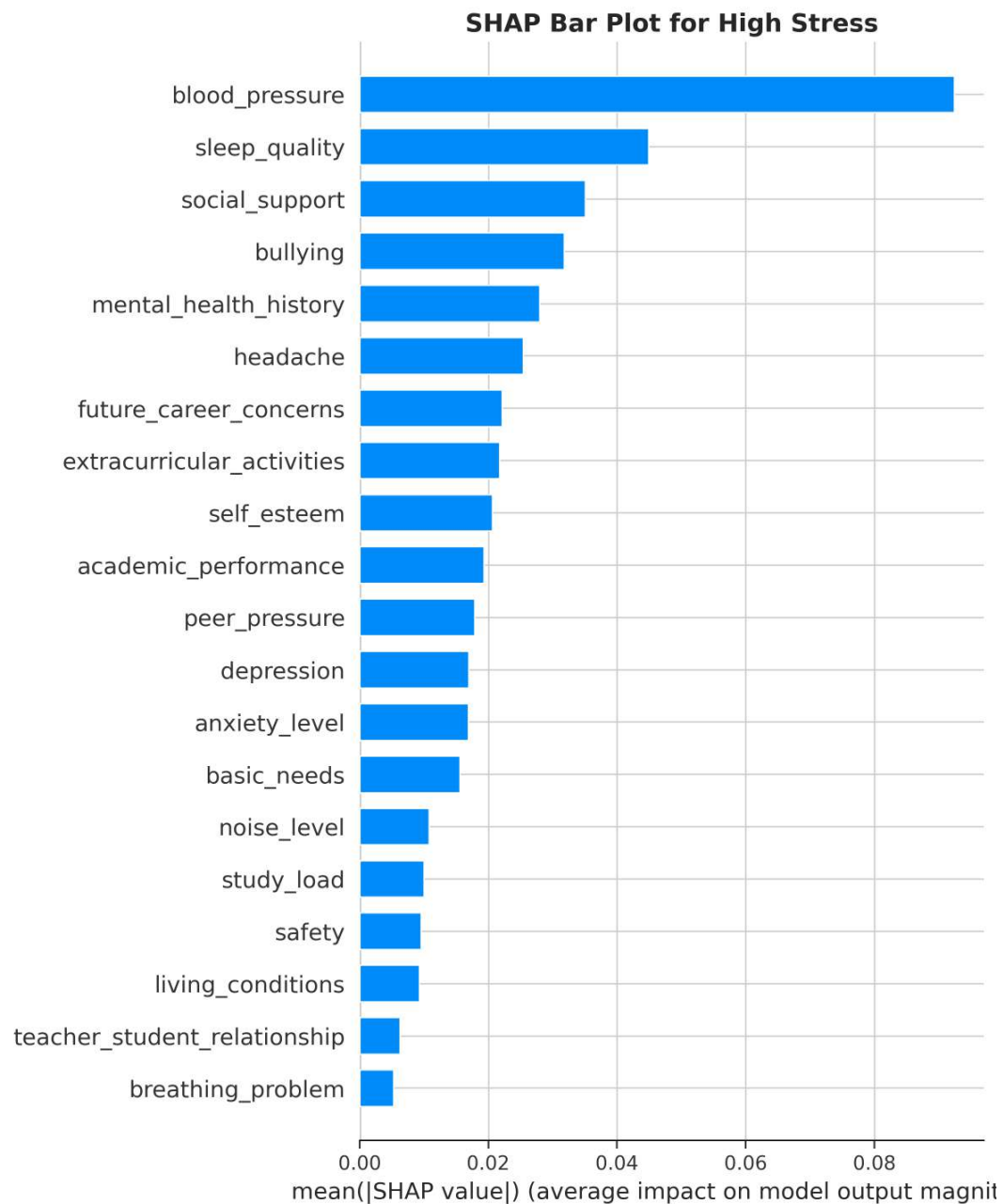


Figure 15: SHAP bar plot showing global feature importance for the final Extra Trees model.

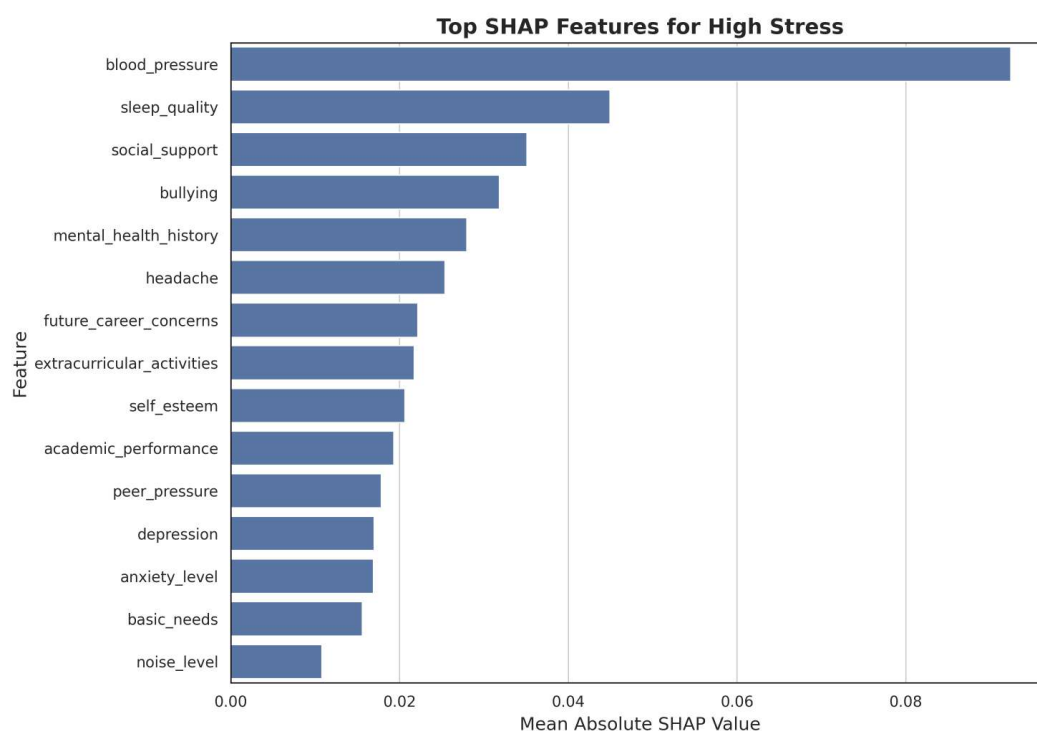


Figure 16: Mean absolute SHAP feature importance plot for the final Extra Trees model.

As seen in Figures 15 and 16, `blood_pressure` emerged as the most influential feature by a significant margin, followed by `headache` and `sleep_quality`, demonstrating that physical and physiological indicators are the strongest overall predictors of stress levels in this dataset.

4.5.2 SHAP Summary Plot

While global feature importance identifies the most influential variables overall, the SHAP summary plot shows how each individual feature specifically contributes to the prediction outcomes for each student, visualizing the direction of effect.

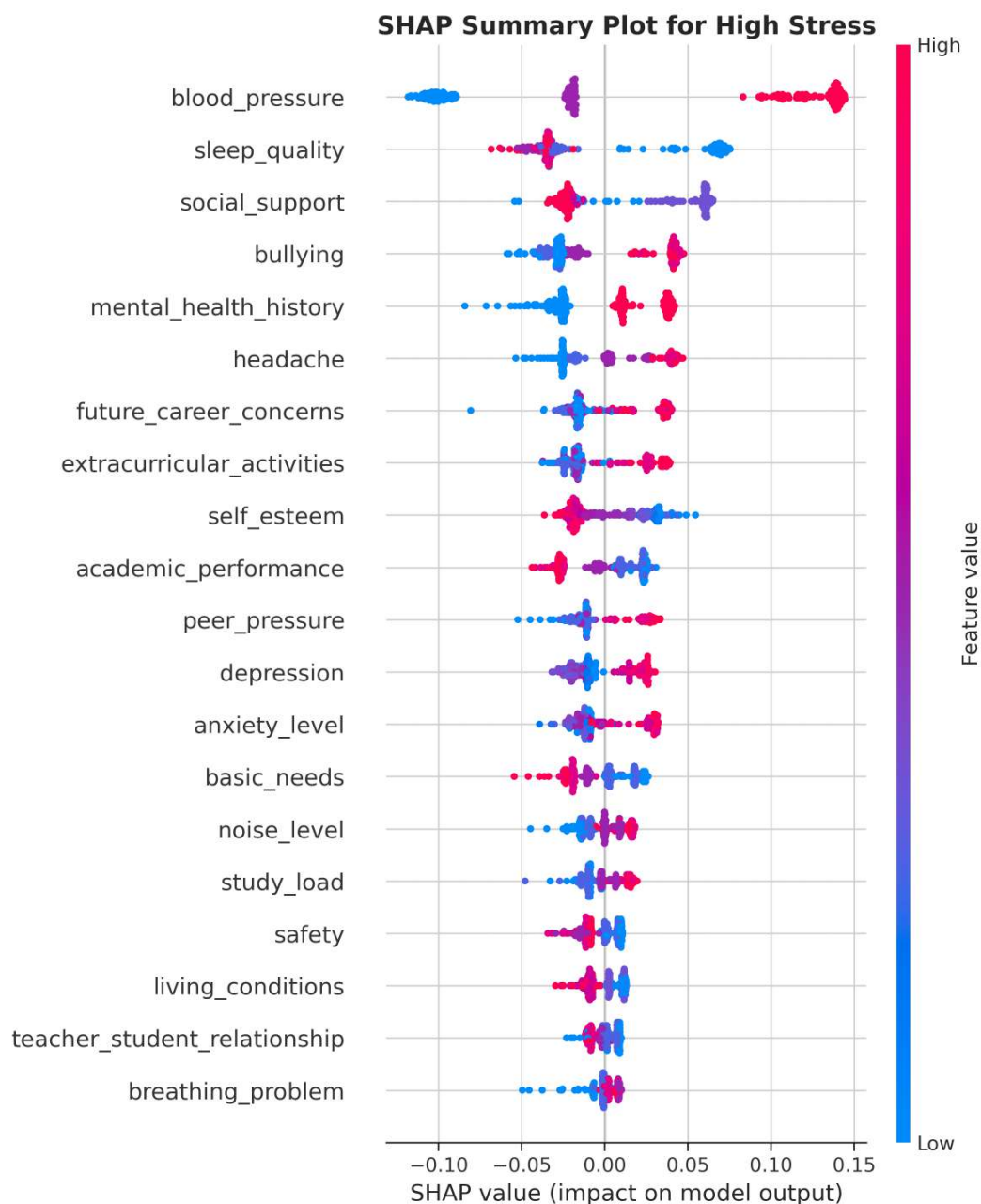


Figure 17: SHAP summary plot showing the specific feature contributions across all test students.

This plot illustrates the impact of each feature across all 220 test students, visually explaining the underlying mechanics of why a student is assigned to the Low, Medium, or High Stress category. This level of explainability ensures the system is interpretable and suitable for practical student monitoring and counselling support.

4.6 Multi-Dimensional Stress Profiles and Student Reports

The defining capability of the framework is its ability to generate personalized, student-level reports by aggregating SHAP values into the five predefined stress dimensions. For each student, the dimension-level SHAP contributions are computed and the dimension with the highest contribution is identified as the Dominant Stress Dimension.

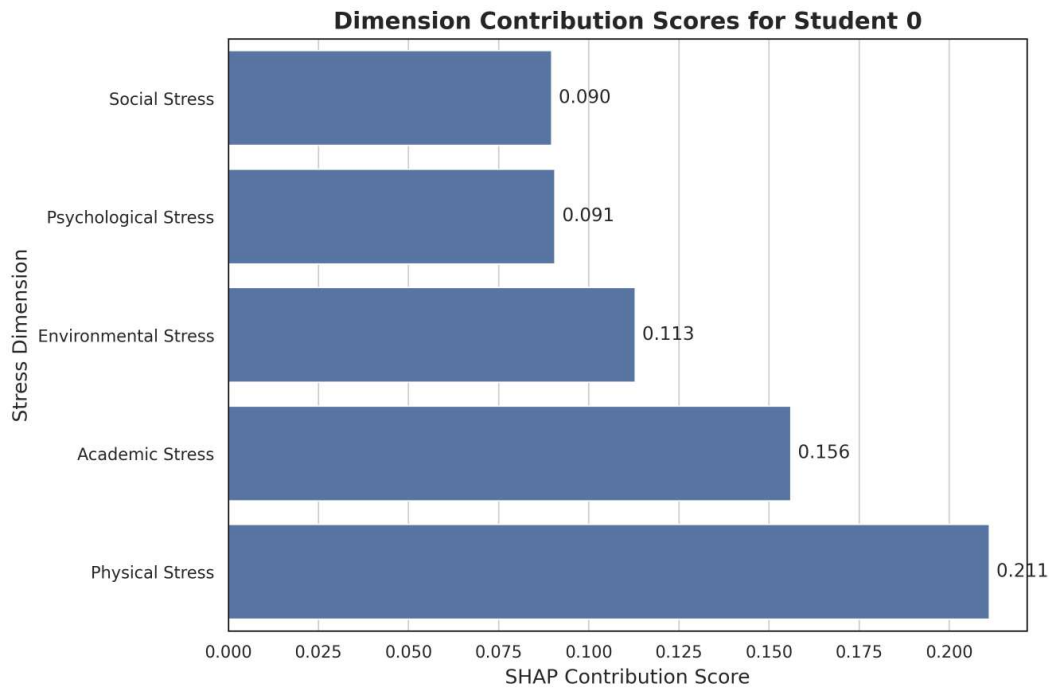


Figure 18: Student-level dimension contribution scores showing the distribution of SHAP contributions across the five stress dimensions.

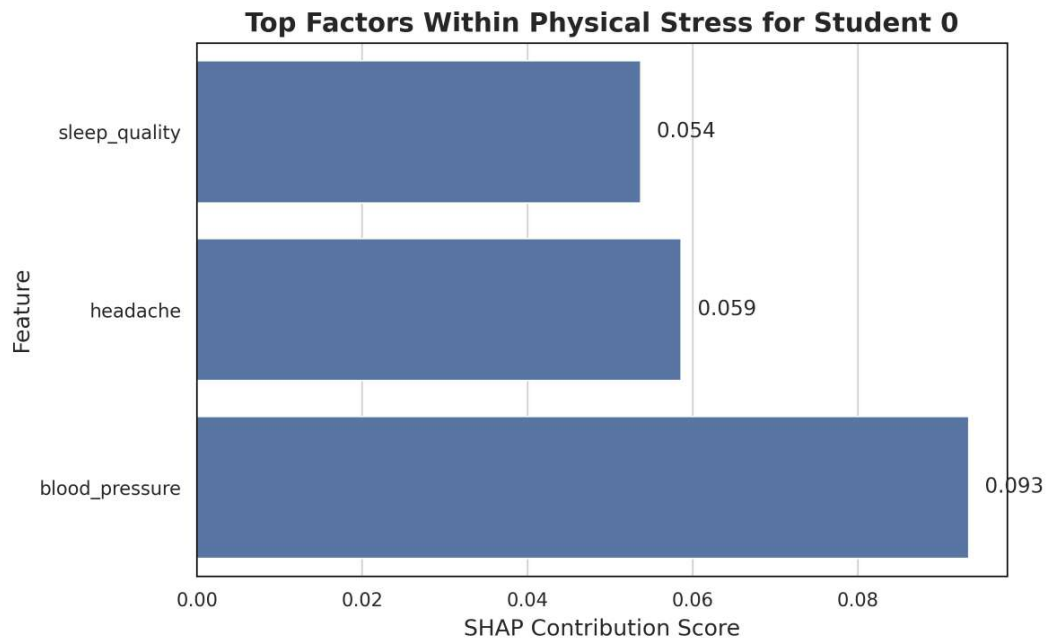


Figure 19: Top factors within the dominant stress dimension, restricted to features belonging strictly to that dimension.

To demonstrate the system's output, consider two example students. For a student predicted with **Low Stress**, the report correctly avoids labelling a 'problem' dimension:

Predicted Stress Level: Low

Dominant Low-Stress Explanation Dimension: Physical Stress

Top Factors Within Dominant Dimension: (1) blood_pressure (2) headache (3) sleep_quality

Recommended Interventions: No intervention required. Monitoring only.

For a student predicted with **High Stress**, the system generates targeted interventions:

Predicted Stress Level: High

Dominant Stress Dimension: Academic Stress

Top Factors Within Dominant Dimension: (1) academic_performance (2) study_load (3) future_career_concerns

Recommended Interventions: Weekly Study Planning; Academic Advisor Meeting; Time Management Training

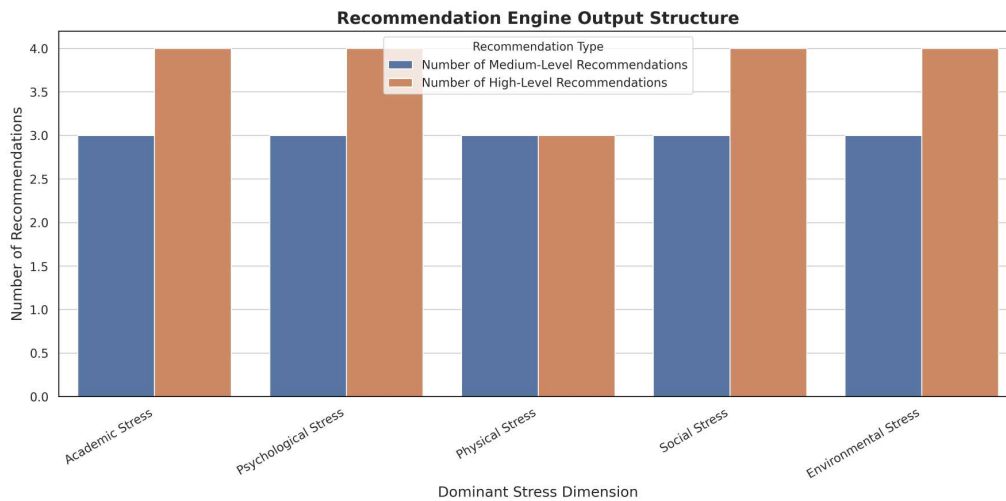


Figure 20: Recommendation engine structure showing the rule-based logic connecting predicted stress level and dominant dimension to personalized interventions.

5 Conclusion

5.1 Summary of Work

This project produced a complete end-to-end machine learning pipeline for student stress analysis. Starting from 1,100 student records with 20 features and a three-class target, we carried out exploratory data analysis, applied leakage-aware preprocessing, trained and compared twelve classifiers, tuned the top three with RandomizedSearchCV, explained the final model using SHAP, detected the dominant stress dimension for each student, restricted the displayed factors to that dimension, and connected the output to a rule-based recommendation engine. The final tuned Extra Trees classifier achieved **89.55% accuracy** and 89.52% weighted F1-score on the 220-record test set.

5.2 Final Decision and Suggestions

The tuned Extra Trees classifier is our recommended model for this pipeline. It delivered the best post-tuning accuracy (89.55%) and the most balanced per-class performance (recall above 85% for all three classes), and it is fully compatible with SHAP’s TreeExplainer for efficient exact computation. Its optimal configuration — 108 trees, maximum depth 17, minimum samples per split 8, minimum samples per leaf 1, Gini criterion — was found by RandomizedSearchCV.

Our practical suggestions for teams pursuing similar work are: (1) Do not assume hyperparameter tuning will always improve a model — as demonstrated here with LightGBM and XGBoost, it can reduce performance with a limited search budget. (2) Use nested cross-validation

rather than reusing the same test set for both model selection and final evaluation. (3) Validate recommendation outputs with qualified counsellors before exposing them to students or staff.

5.3 Limitations

This study has several important limitations. First, the dataset features are self-reported, integer-coded survey responses — `blood_pressure` is an ordinal variable (1, 2, 3) not a clinical measurement. Second, the same 220-record test set was reused to evaluate all twelve baseline models, shortlist the top three, and select the final model, inflating the apparent reliability of the 89.55% accuracy. Third, the recommendation engine was designed manually using domain knowledge with no intervention-outcome data to verify effectiveness. Fourth, SHAP values are associational, not causal.

5.4 Future Research Directions

Building on the findings and acknowledged limitations: (1) Replace `RandomizedSearchCV` with Bayesian optimization (Optuna or Hyperopt) for a more targeted search. (2) Apply stacking or voting ensembles on the top models to close the accuracy gap with literature benchmarks. (3) Conduct an ablation study removing `blood_pressure` to understand its true influence. (4) Collect longitudinal data to enable early-warning prediction across a semester. (5) Deploy the pipeline in a real university setting and measure whether recommendations lead to measurable reductions in self-reported stress.

In conclusion, this project demonstrated that an explainable and multi-dimensional machine learning framework for student stress analysis is both technically feasible and practically structured. The accuracy achieved (89.55%) is competitive with prior studies on the same dataset, and the additional value lies in the interpretive and recommendation layer that connects the model's output to actionable guidance.

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